## Basal Ganglia

*Select a blue button to move to that topic*

- Overview of Basal Ganglia structure
- Basal Ganglia circuitry (the Striato-Pallido-Thalamic loop)
- Modulation of inputs to the Basal Ganglia
- Modulation of outputs from the Basal Ganglia
- Summary of extrapyramidal circuitry
- Clinical signs related to the Basal Ganglia
- Effects of drugs on the Nervous System

## Basal Ganglia

*(next)* → Overview of Basal Ganglia structure

Basal Ganglia circuitry (the Striato-Pallido-Thalamic loop)

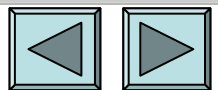
Modulation of inputs to the Basal Ganglia

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Extrapyramidal Motor System

# Basal Ganglia

The basal ganglia perform two functions:

1. Initiate movements

1. Regulate stereotypic movements

*Select a button to view that example*

#1

#2a

#2b

# Initiate movements



Play Movie

The basal ganglia govern the initiation changes in motor behavior - meaning both *starting* AND *stopping* of motor actions. Sipping from a cup requires proper initiation or stopping of multiple, sequential actions.



*(return…)*

# Stereotypic movements #1



Play Movie

Stereotyped movements, like other habits, often are context dependent. The cerebellum governs the movement itself, but the initiation of the motor pattern in context is governed by the basal ganglia. The basal ganglia have been implicated in the expression of obsessive-compulsive disorders.

# Stereotypic movements #2

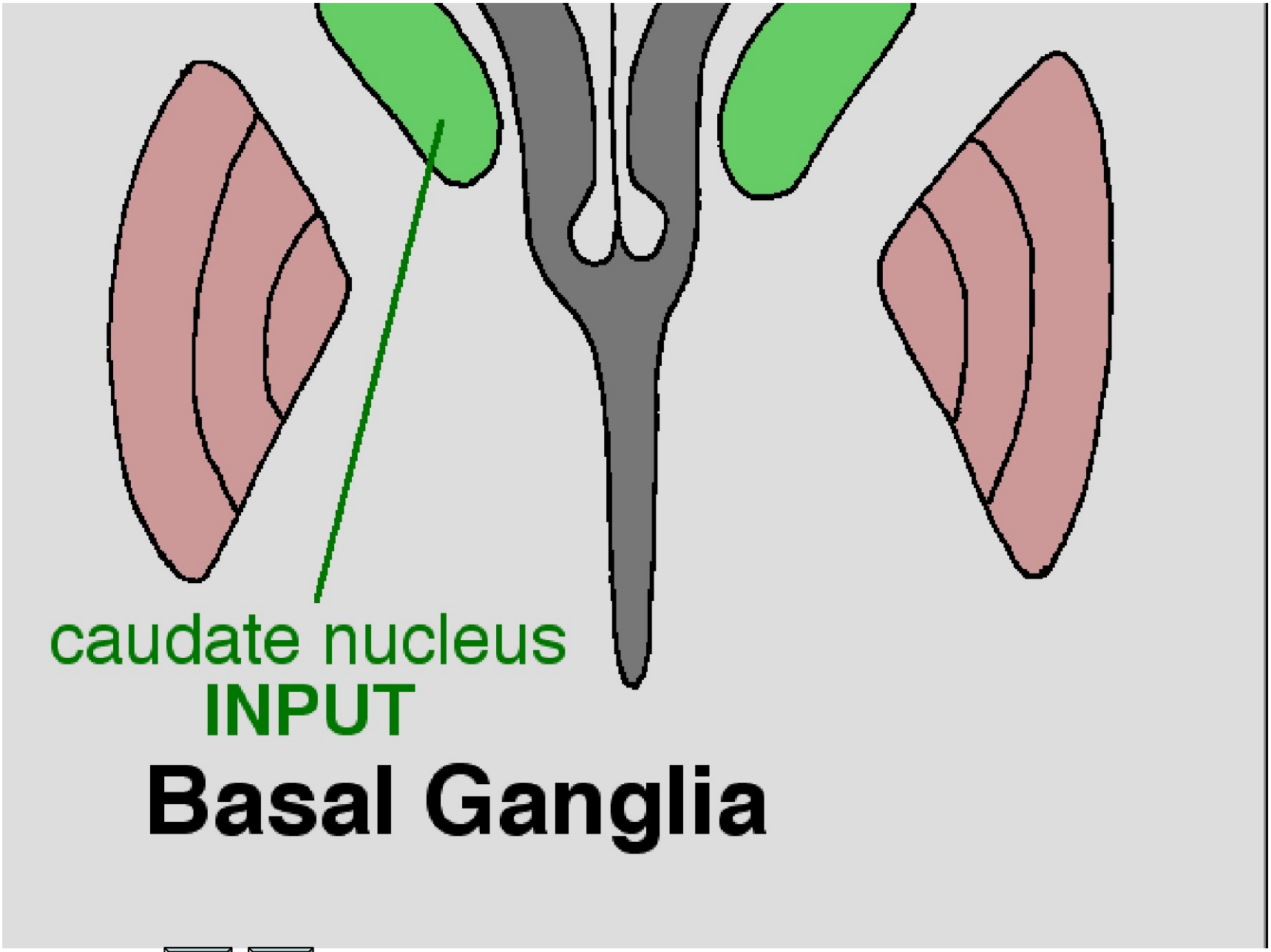


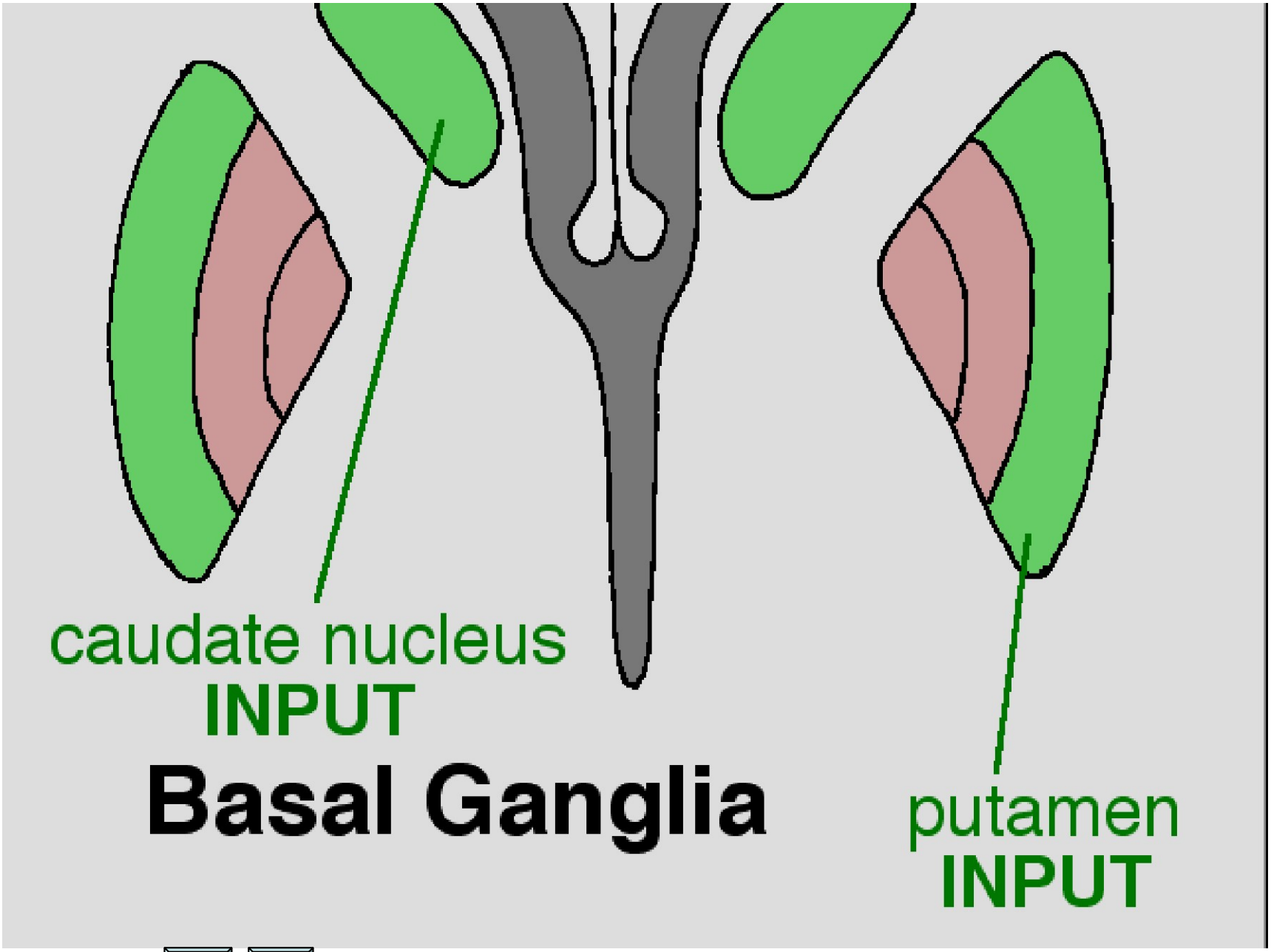
Play Movie

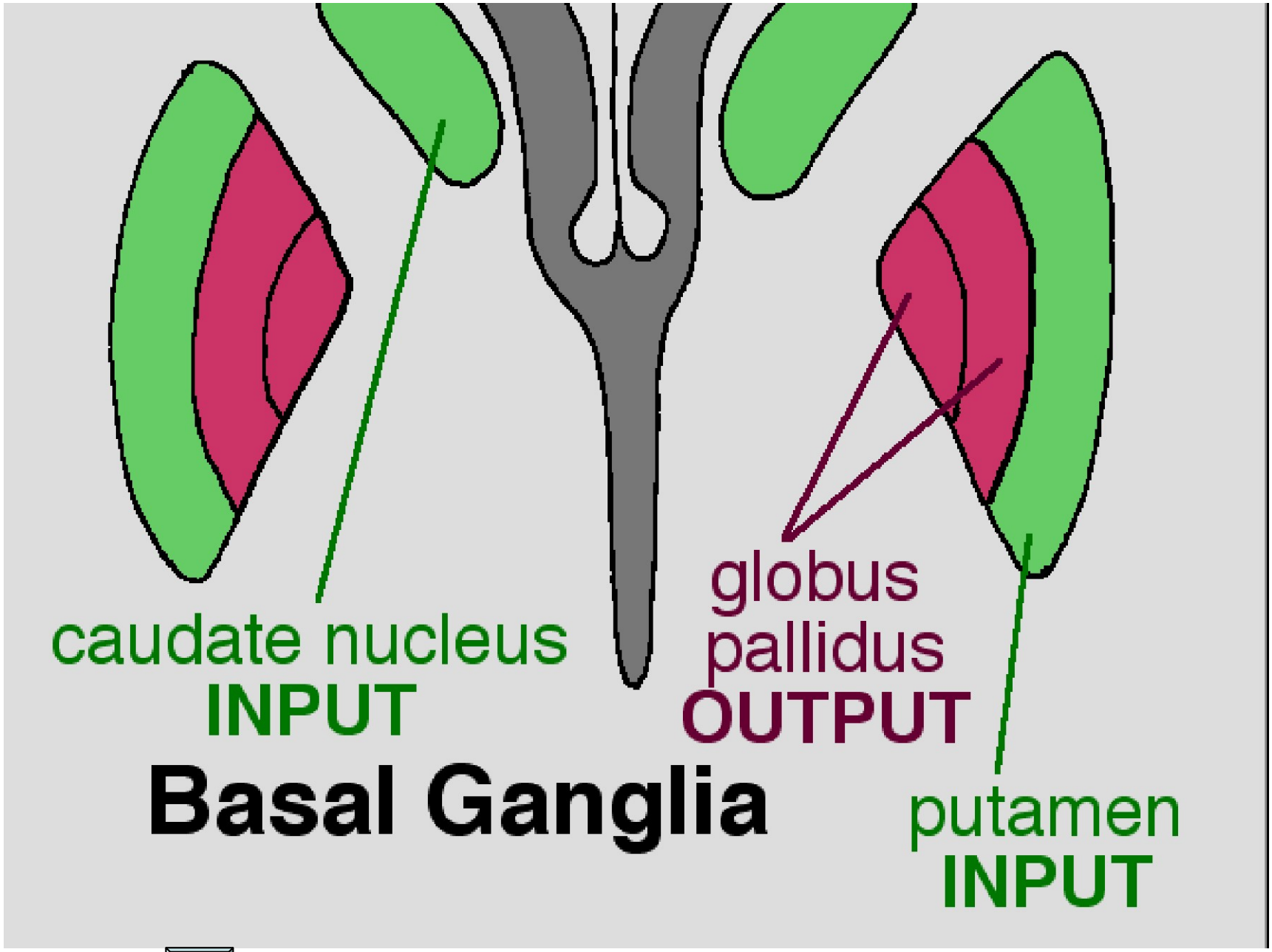
Not all stereotyped behaviors are rhythmic, and the motor pattern can be difficult to halt once it has begun.  Have you ever tried to stifle a yawn?



*(return…)*


caudate nucleus
INPUT
Basal Ganglia


caudate nucleus
INPUT
Basal Ganglia
putamen
INPUT


caudate nucleus
INPUT
globus
pallidus
OUTPUT
Basal Ganglia
putamen
INPUT

## Basal Ganglia

Overview of Basal Ganglia structure

*(next)* → Basal Ganglia circuitry (the Striato-Pallido-Thalamic loop)

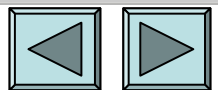
Modulation of inputs to the Basal Ganglia

Modulation of outputs from the Basal Ganglia

Summary of extrapyramidal circuitry

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# Extrapyramidal Circuitry

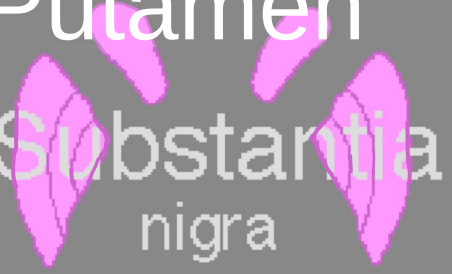
The basal ganglia form an internal motor circuit …

Cortex

Caudate

Putamen

Globus Pallidus



Substantia nigra

Ventral Anterior N.

Cerebellum

Ventral Lateral N.

Subthalamus

Intralaminar N.

Extrapyramidal Circuitry

# Striato-Pallido-Thalamic Loop




Globus Pallidus

The cortex receives motor planning information, then passes that information to the caudate & putamen, which govern timing of events

Substantia nigra

Ventral Anterior N.

Cerebellum

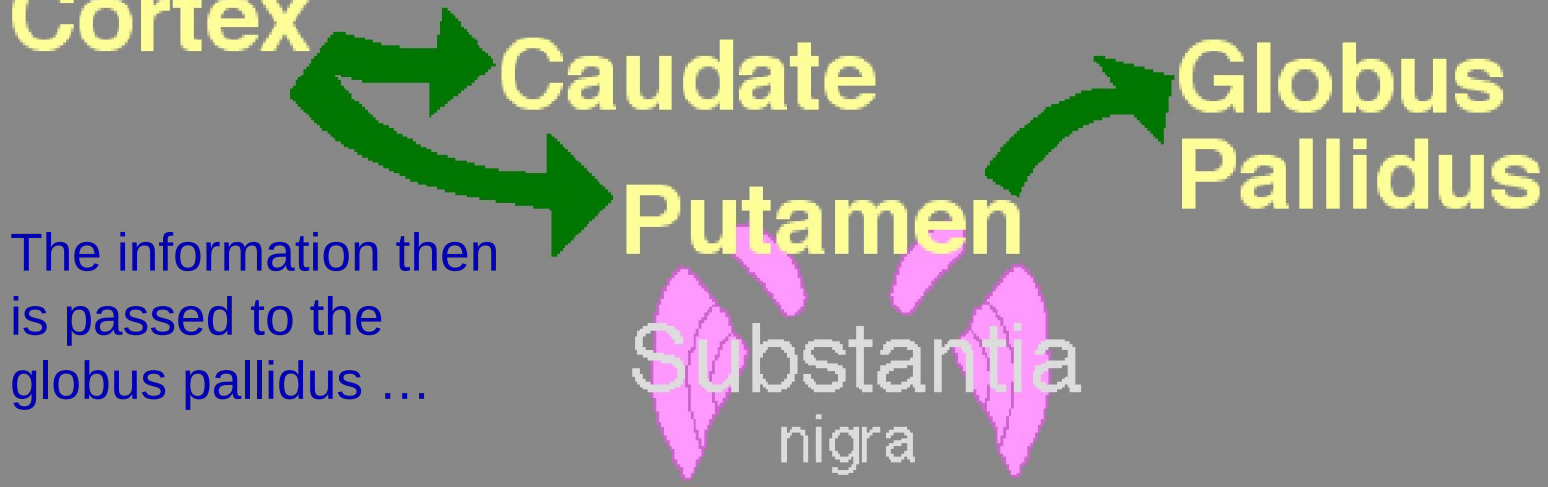
Ventral Lateral N.

Subthalamus

Intralaminar N.

Extrapyramidal Circuitry

# Striato-Pallido-Thalamic Loop




The information then is passed to the globus pallidus …

Ventral Anterior N.

Ventral Lateral N.

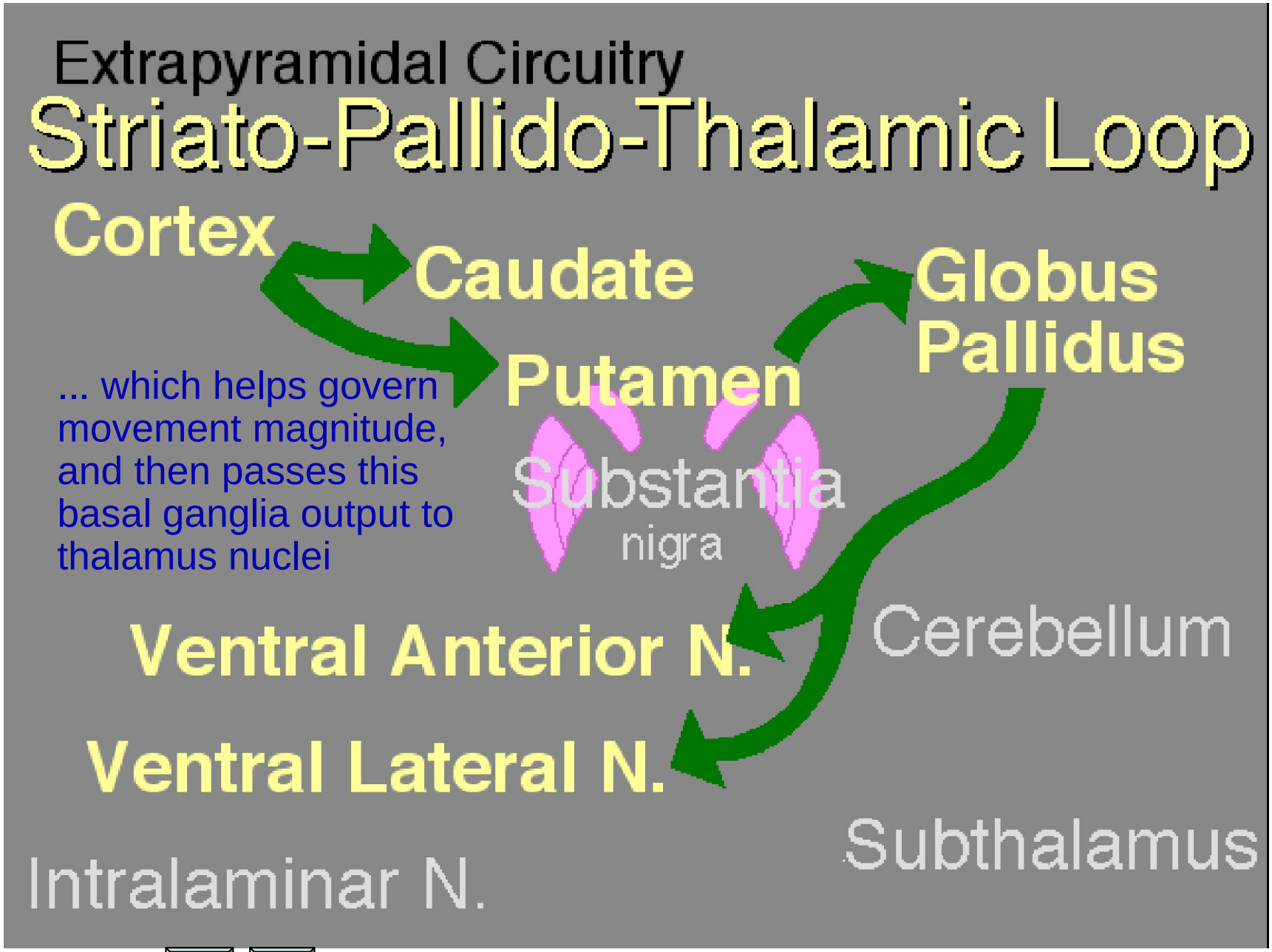
Intralaminar N.

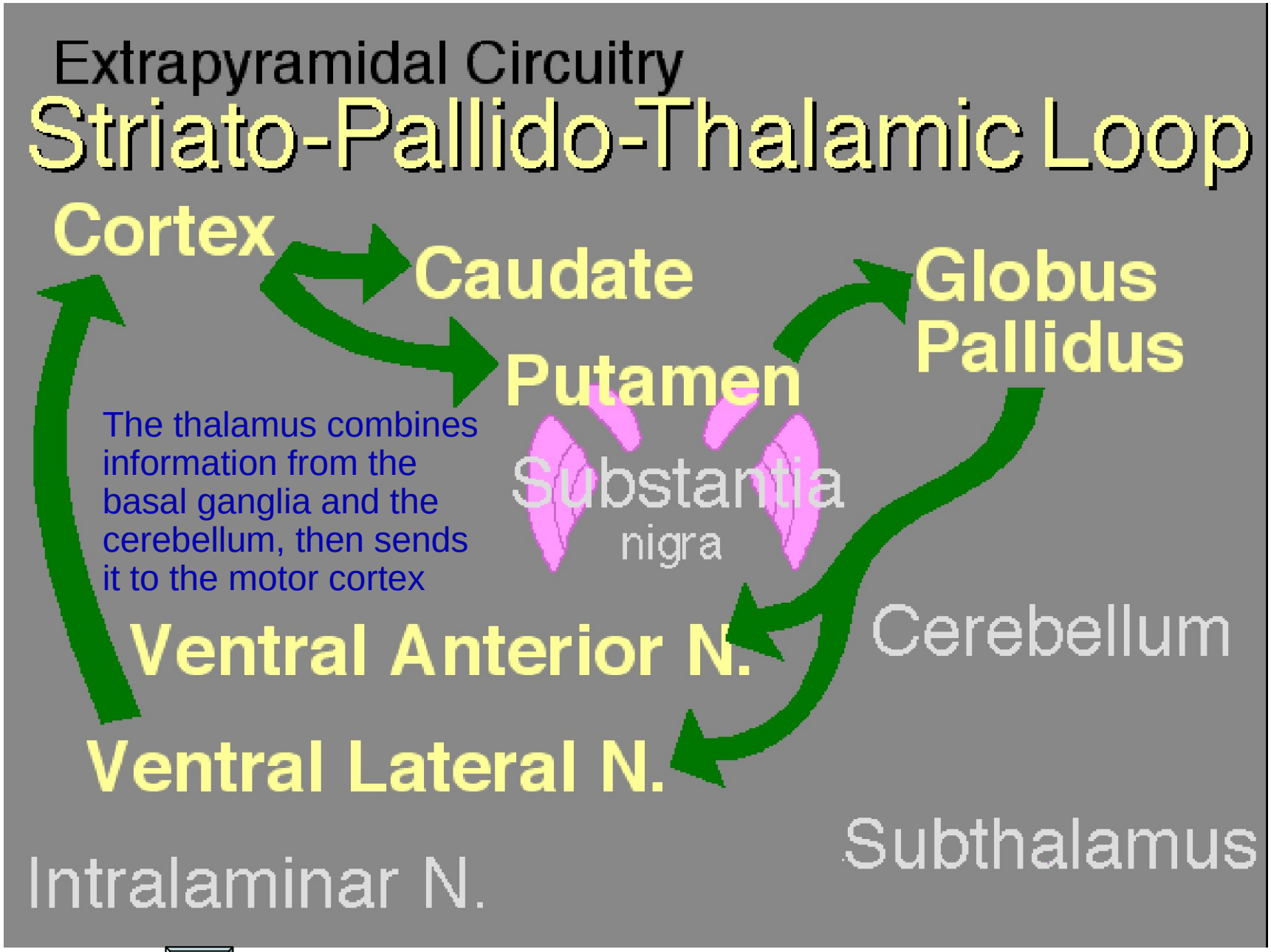
Cerebellum

Subthalamus

Extrapyramidal Circuitry

# Striato-Pallido-Thalamic Loop




Extrapyramidal Circuitry
Striato-Pallido-Thalamic Loop
Cortex
Caudate
Putamen
Globus Pallidus
The thalamus combines information from the basal ganglia and the cerebellum, then sends it to the motor cortex
Substantia nigra
Ventral Anterior N.
Cerebellum
Ventral Lateral N.
Subthalamus
Intralaminar N.

## Basal Ganglia

Overview of Basal Ganglia structure

Basal Ganglia circuitry (the Striato-Pallido-Thalamic loop)

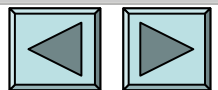
*(next)* → Modulation of inputs to the Basal Ganglia

Modulation of outputs from the Basal Ganglia

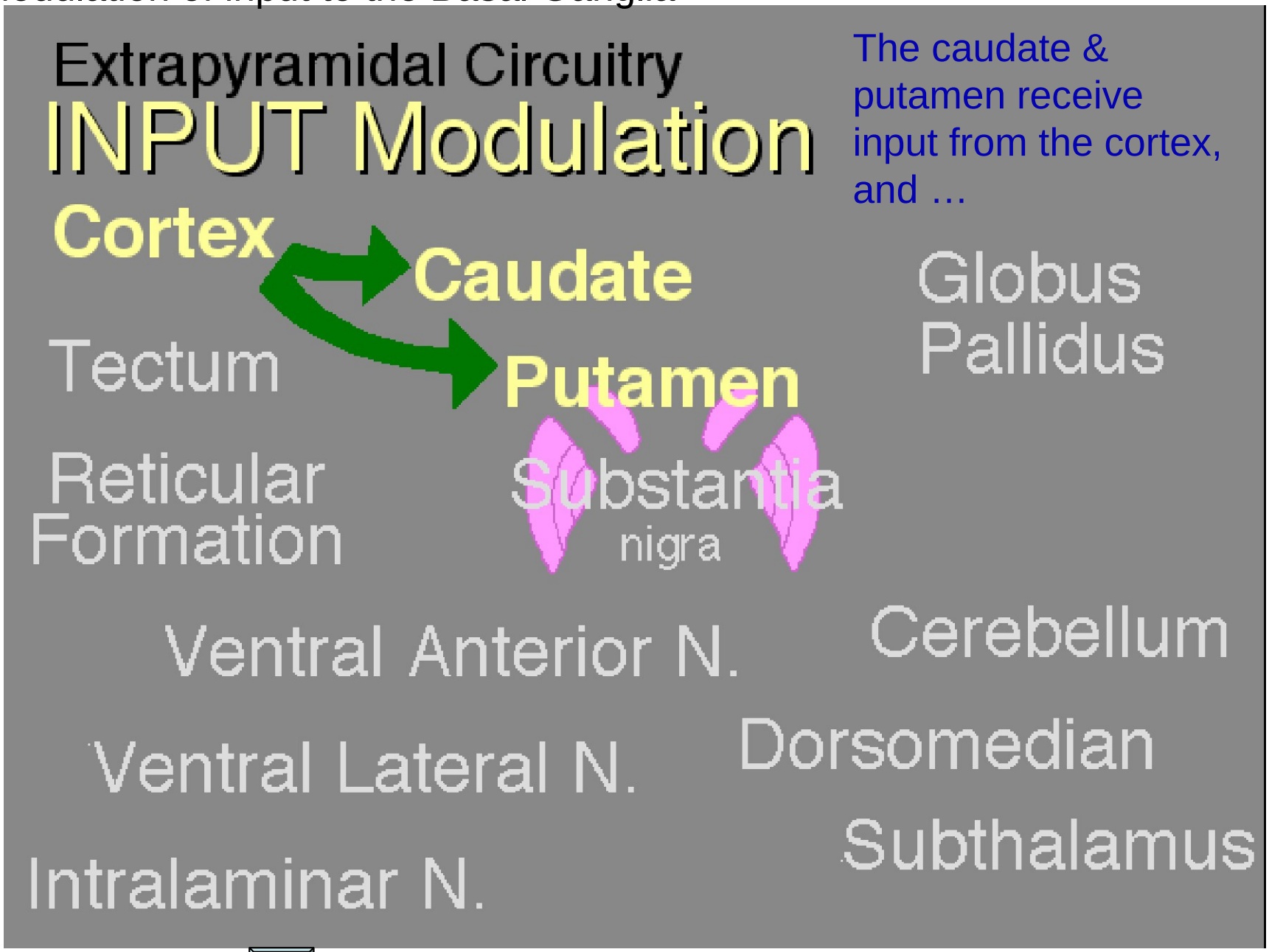
Summary of extrapyramidal circuitry

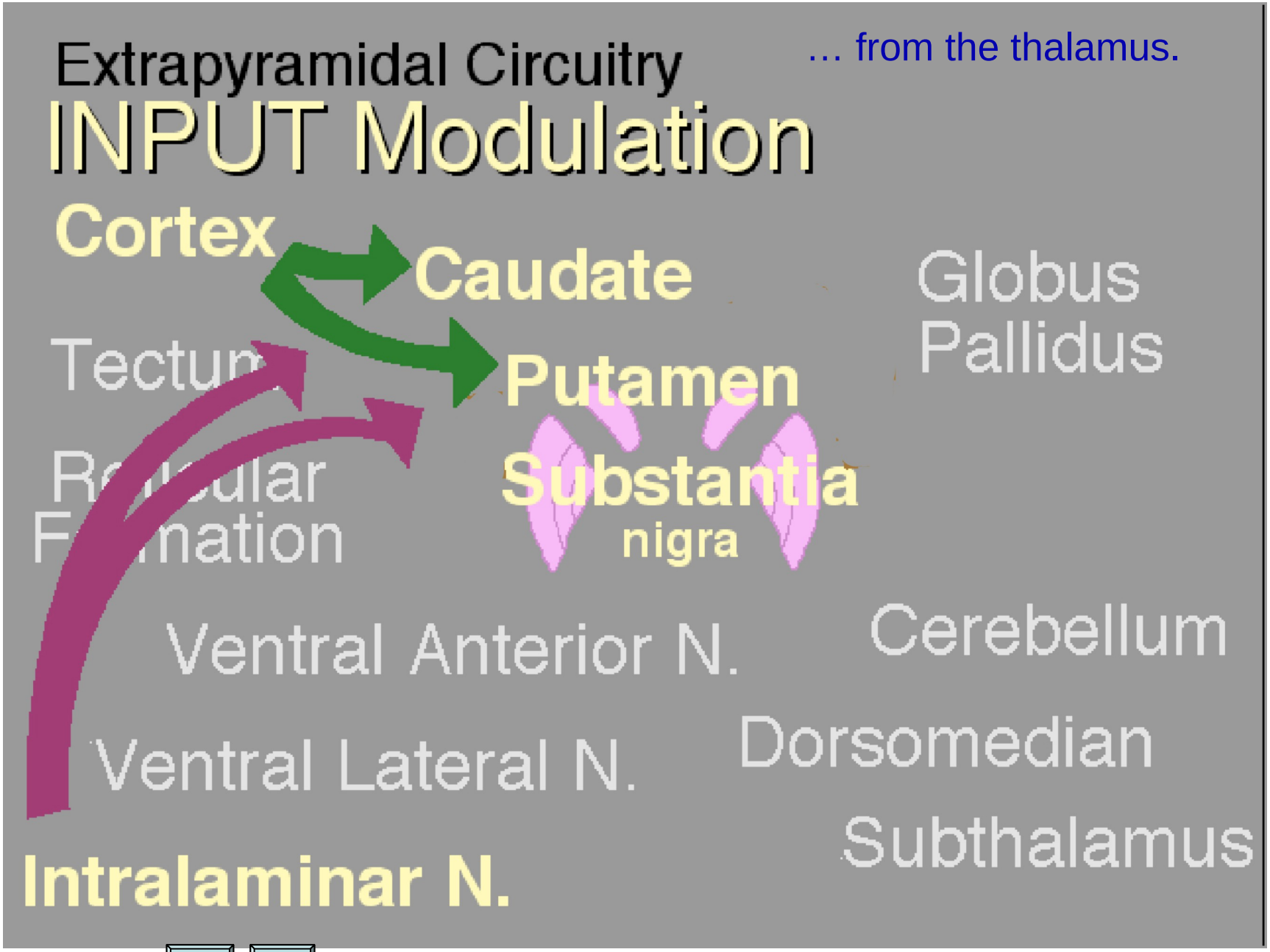
Clinical signs related to the Basal Ganglia

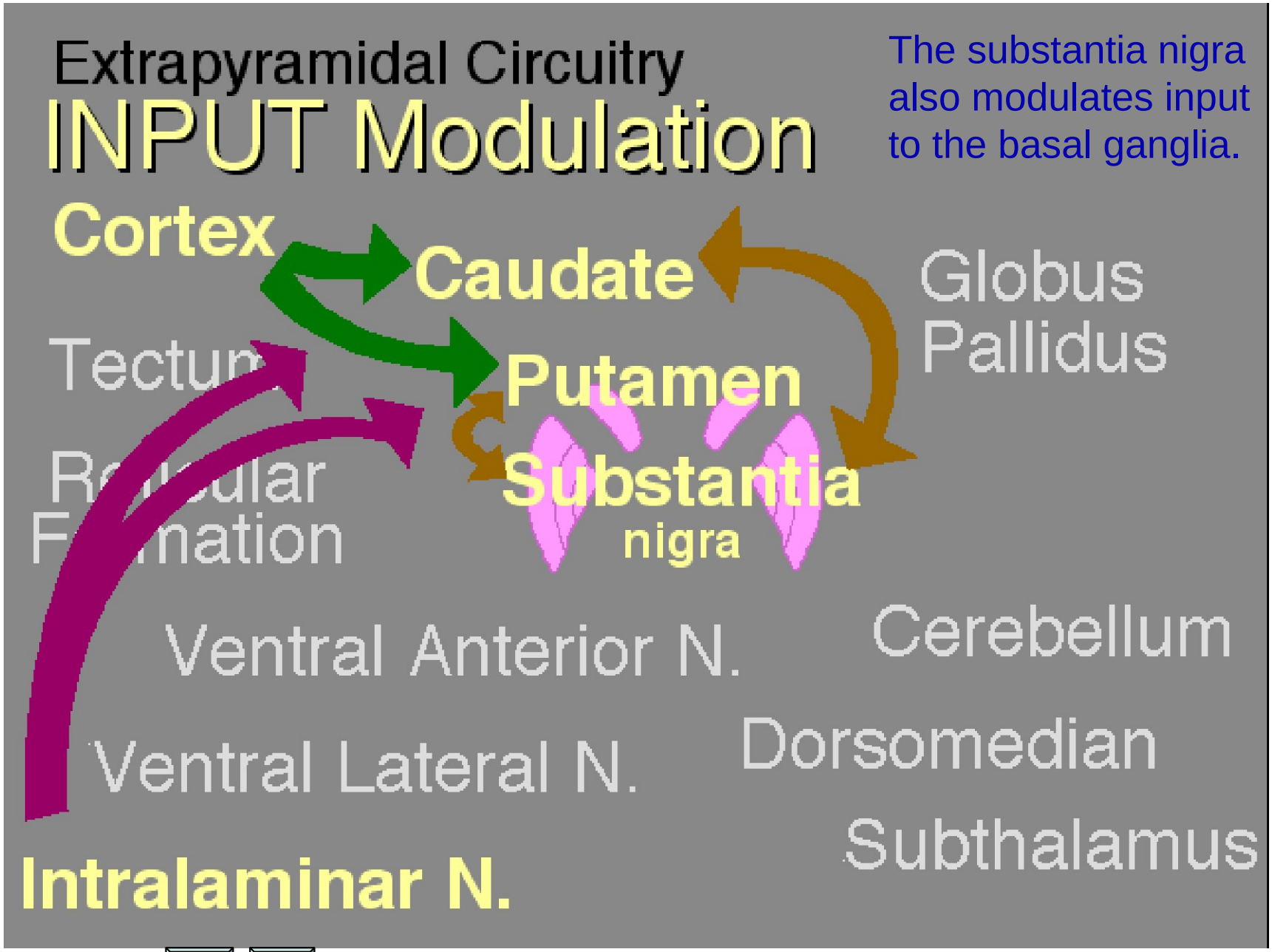
Effects of drugs on the Nervous System



Modulation of input to the Basal Ganglia




Extrapyramidal Circuitry
… from the thalamus.
INPUT Modulation
Cortex
Caudate
Putamen
Globus
Pallidus
Tectum
Substantia
nigra
Ventral Anterior N.
Cerebellum
Ventral Lateral N.
Dorsomedian
Subthalamus
Intralaminar N.


Extrapyramidal Circuitry
INPUT Modulation
The substantia nigra also modulates input to the basal ganglia.
Cortex
Caudate
Putamen
Substantia nigra
Globus Pallidus
Tectum
Reticular Formation
Ventral Anterior N.
Ventral Lateral N.
Intralaminar N.
Cerebellum
Dorsomedian
Subthalamus

Extrapyramidal Circuitry

# INPUT Modulation




Reciprocal connections with the caudate & putamen allow exitatory inputs from the substantia nigra to modulate the amount and type of output sent to the globus pallidus. Dopamine is the neurotransmitter used by these substantia nigra pathways.

Extrapyramidal Circuitry

# INPUT Modulation




When the substantia nigra isn't working properly, input to the basal ganglia isn't modulated properly, and the globus pallidus receive progressvely less information. Without this information, the initiation of movement (*i.e.,* timing) message is less effective and the person's movements progressively become slower (*i.e.,* bradykinesia).

Extrapyramidal Circuitry

# INPUT Modulation




**Parkinson's disease** is related to a deterioration of the substantia nigra and globus pallidus, and is characterized by resting tremors and bradykinesia.

## Basal Ganglia

Overview of Basal Ganglia structure

Basal Ganglia circuitry (the Striato-Pallido-Thalamic loop)

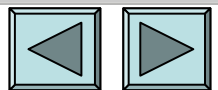
Modulation of inputs to the Basal Ganglia

*(next)* → Modulation of outputs from the Basal Ganglia

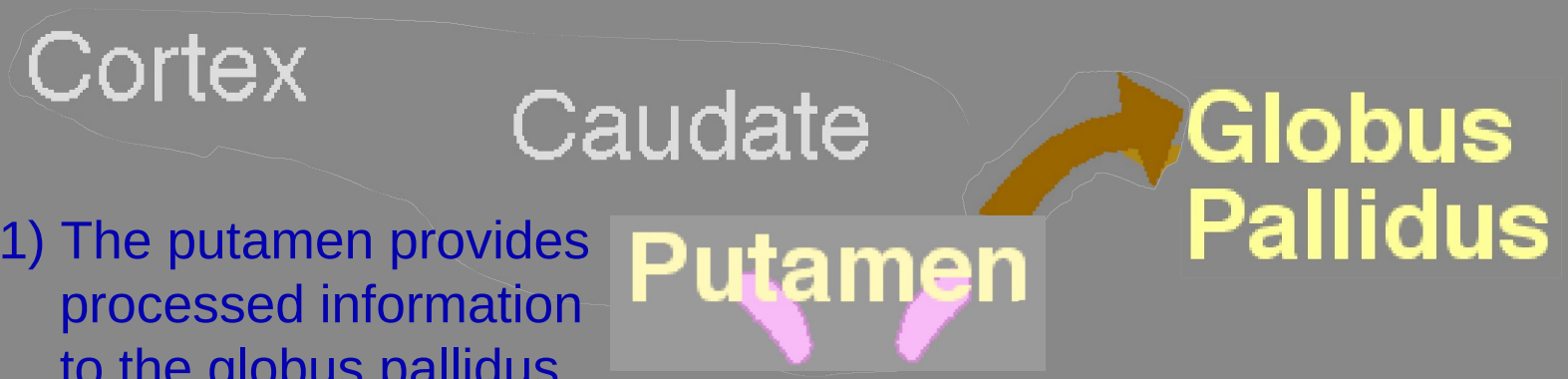
Summary of extrapyramidal circuitry

Clinical signs related to the Basal Ganglia

Effects of drugs on the Nervous System

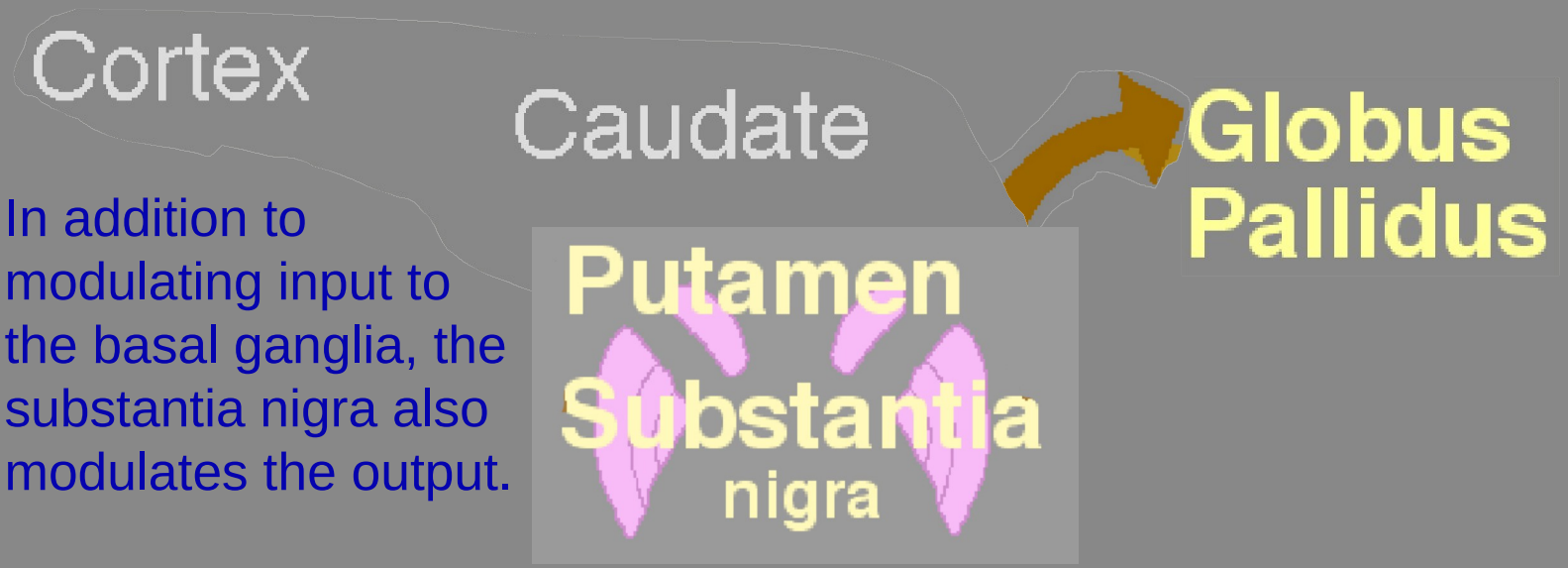


# OUTPUT Modulation - part 1



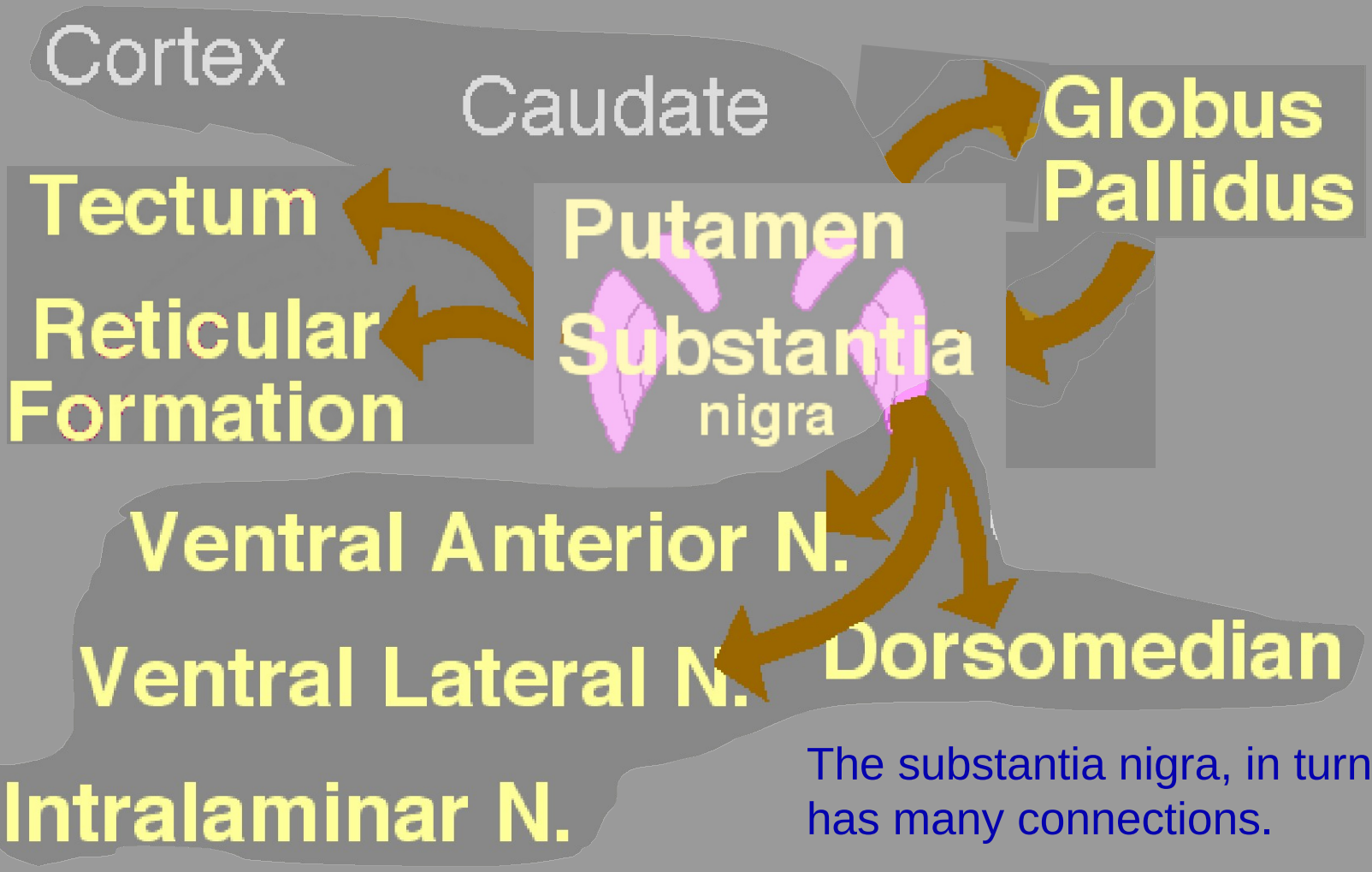

1) The putamen provides processed information to the globus pallidus.

# OUTPUT Modulation - part 1




In addition to modulating input to the basal ganglia, the substantia nigra also modulates the output.

# OUTPUT Modulation - part 1




The substantia nigra, in turn, has many connections.

Extrapyramidal Circuitry

# OUTPUT Modulation - part 2

Cortex

Caudate

Putamen

Substantia nigra

**Globus Pallidus**

2) The subthalamus plays a role in modulating output from the basal ganglia

Ventral Anterior N.

Ventral Lateral N.

Intralaminar N.

Cerebellum

**Subthalamus**

Extrapyramidal Circuitry

# OUTPUT Modulation - part 2

Cortex

Caudate

Putamen

Substantia nigra

Globus Pallidus

Deterioration of the subthalamus results in the ballisms, or explosive movements occurring periodically, that characterize Huntington's disease.

Ventral Anterior N.

Ventral Lateral N.

Intralaminar N.

Cerebellum

Subthalamus



## Basal Ganglia

Overview of Basal Ganglia structure

Basal Ganglia circuitry (the Striato-Pallido-Thalamic loop)

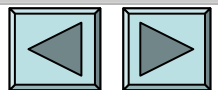
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Modulation of outputs from the Basal Ganglia

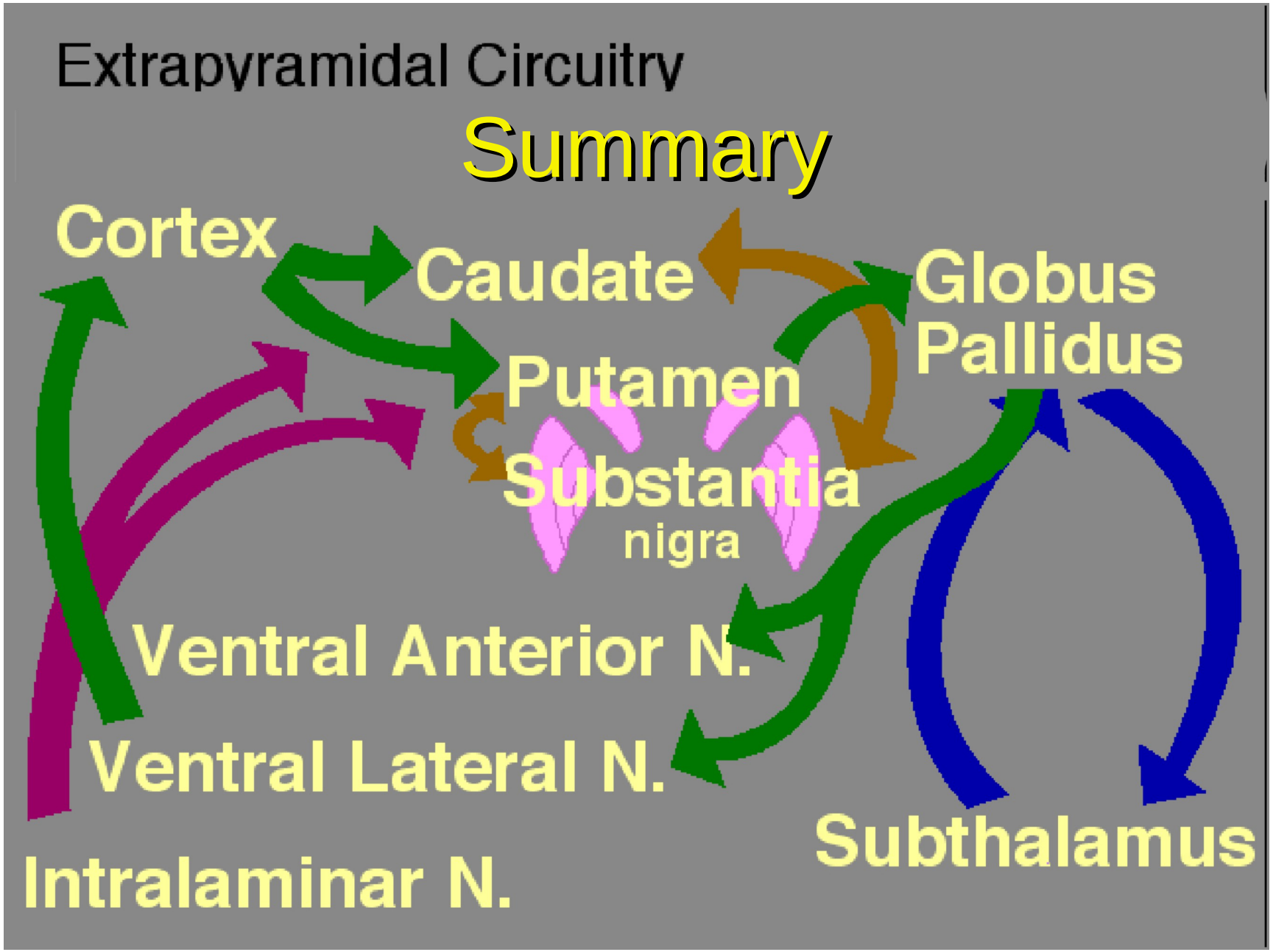
*(next)* → Summary of extrapyramidal circuitry

Clinical signs related to the Basal Ganglia

Effects of drugs on the Nervous System



# Summary


Extrapyramidal Circuitry
Cortex
Caudate
Globus Pallidus
Putamen
Substantia nigra
Ventral Anterior N.
Ventral Lateral N.
Intralaminar N.
Subthalamus

**Basal Ganglia**

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# Clinical signs related to Basal Ganglia dysfunction

## Parkinson's disease

resting tremor . . . . . #1
poverty of movement . . . . . #2
bradykinesia . . . . . #3
stooped posture . . . . . #4
cogwheel rigidity . . . . . #5
masked face . . . . . #6

*Select a blue button to view the video*

## Chorea . . . . . #7

## Choreaform movements . . #8

## Tics . . . . . #9

## Ballisms . . . . . #10a #10b

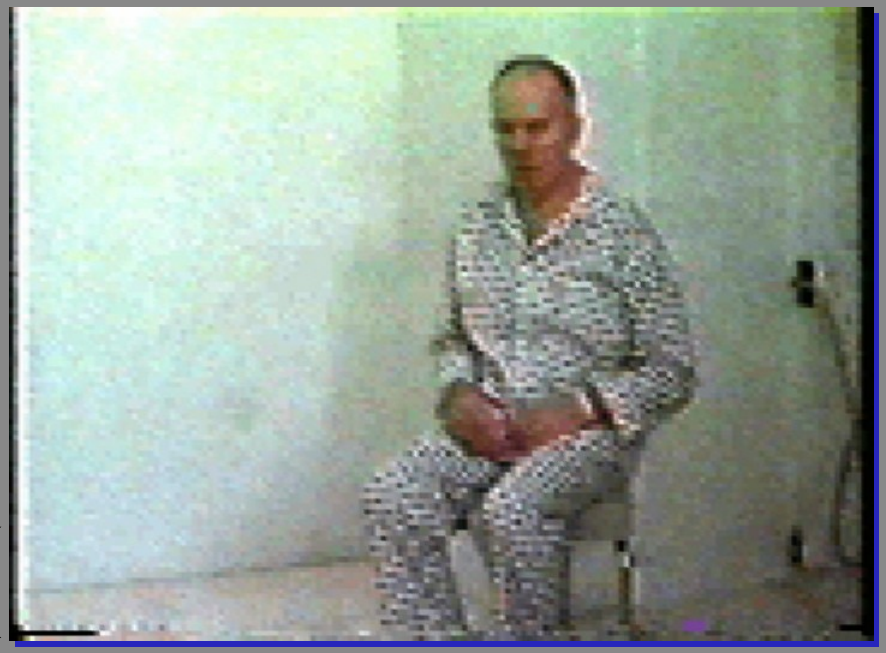
*(YouTube's website also can be a good source of example videos…)*



Basal Ganglia menu

# Resting tremor

Play Movie



An hallmark feature of Parkinson's disease is the resting tremor - involuntary, oscillating movements at rest, often of the hands or head and neck.

*It is not a resting tremor if it occurs during execution of a movement!*

# Poverty of movement



Play Movie

Limited or no movement - often described by the person as becoming “stuck” in position - also is characteristic of Parkinson’s disease. Remember that the basal ganglia are responsible for initiating a change in movement patterns. Here, the individual is attempting to rise from a chair - a complex motor sequence where timing of the component movements is critical.

# Bradykinesia

Play Movie



“Slowed movement” is another feature of Parkinson’s disease, and is particularly noticeable with walking - a motor behavior where the center of gravity shifts forward, and placing each foot forward alternately is what keeps you from falling.  Slowed movements allows the center of gravity to shift too far forward, too rapidly.

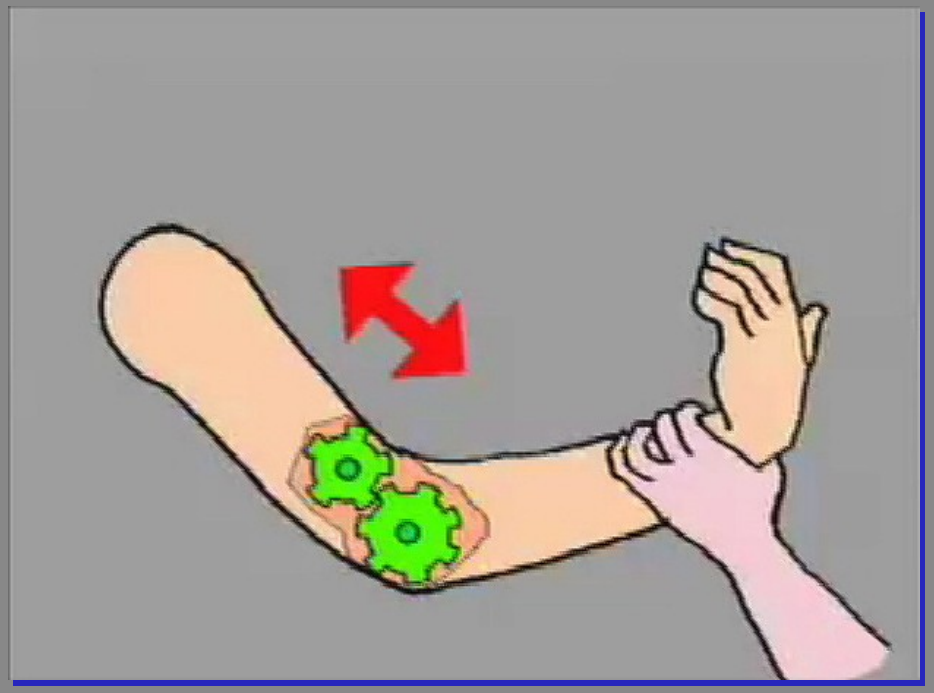
# Stooped posture



Play Movie

A biomechanical feature of many disorders, but characteristic of Parkinson's disease when seen in combination with other symptoms. Here, note the resting tremor, bradykinesia, and - despite the poor quality image - a masked facial expression.

# Cogwheel rigidity



Play Movie

An animation of the concept, meant to illustrate the lack of smooth movements at the joint. The feeling is similar to that of a cogwheel:

*“click - stop - click - stop...”*

# Masked facial expression












*"The Phantom of the Opera" Warner Brothers, 2004 Gerard Butler as the Phantom*

Conveying emotion through facial expression requires initiation of motor behavior. Without these behaviors, the person may have a "masked" facial expression. Be aware, however, that emotional capacity is undiminished even if the emotion can't be expressed!

# Chorea



Play Movie

Huntington's disease and other choreas typically are disorders in modulation of the output from the basal ganglia. Ongoing, smoothly flowing movements are typical - but the constant movement consumes calories rapidly. The movements do not occur during sleep.

A different  form of chorea (now called Sydenham's chorea) was once termed St. Vitus' dance due to the frenzied limb movements and contorted facial expressions.  This form of chorea can appear after rheumatic fever, but also may co-occur with obsessive-compulsive disorder - which also is associated with basal ganglia dysfunction.

# Choreaform movements

Play Movie



Other intrusive movements may appear similar to those of Huntington's chorea and Sydenham's chorea, but do not arise from the same causes (genetic source or bacterial infection). These movements are termed "choreaform" and they can be significant barrier to function. This individual has been asked to touch her nose with each index. Unlike true choreas, her impairment primarily is unilateral (right side).

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*(next)* ⟶ Effects of drugs on the Nervous System

# L-dopa therapy



Play Movie

**L-dopa** is used to treat Parkinson's disease. While very effective in the earlier stages of the disorder, symptoms continue to develop and larger doses of drug are required. Eventually the side effects if high doses of L-dopa will impair function and diminish quality of life. This is an example of a drug regime that allows the symptoms of a disorder to be managed - at least for a time - but is NOT a cure for the disorder. This example also shows three stages of drug action - prior to treatment (poverty of movement; resting tremor; stooped posture; shuffling gait), soon after administration (intrusive movements), and later on - during the therapeutic phase of the drug's action.

That ends the Basal Ganglia tutorial!